

Chapter

Equation Of A Straight Line

1. Which of the following equation represent a line passing through origin?

(a) $3x - 2y + 5 = 0$

(b) $2x - 3y = 0$

(c) $x = 5$

(d) $y = -6$

SQP 2024

2. Points $A(x, y)$, $B(3, -2)$ and $C(4, -5)$ are collinear. The value of y in terms of x is:

(a) $3x - 11$

(b) $11 - 3x$

(c) $3x - 7$

(d) $7 - 3x$

Main 2024

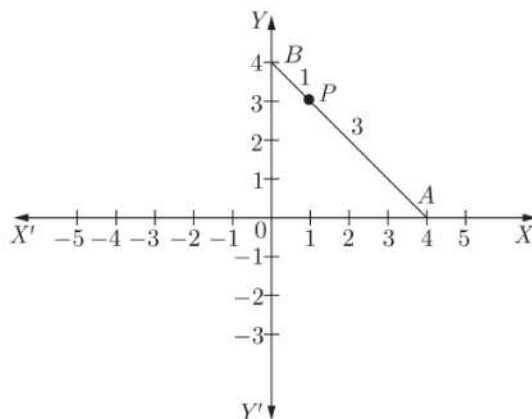
3. ABC is a triangle whose vertices are $A(1, -1)$, $B(0, 4)$ and $C(-6, 4)$. D is the mid point of BC . Find the:

(a) coordinates of D .

(b) equation of the median AD .

Main 2023

4. A and B are two points on the x -axis and y -axis respectively.



(a) Write down the coordinates of A and B .

(b) P is a point on AB such that $AP : PB = 3 : 1$.

Using section formula find the coordinates of point P .

(c) Find the equation of a line passing through P and perpendicular to AB .

Main 2023

5. The equation of a straight line is $5x - 4y - 8 = 0$. What is the y -intercept of the line ?

(a) 2

(b) 8

(c) -2

(d) 4

Comp 2023

6. Find the equation of diagonal BD of rhombus $ABCD$ where $A(4, 5)$ and $C(6, -9)$.

Comp 2023

Solution

1.

Ans: (b) $2x - 3y = 0$

Explanation:

A straight line passes through the origin if the constant term in its equation is zero.

In the equation

$$2x - 3y = 0$$

the constant term is zero.

Hence, this line passes through the origin.

Thus, option (b) is the correct answer.

2.

Ans: (d) $7 - 3x$

Explanation:

Since points $B(3, -2)$ and $C(4, -5)$ lie on the same straight line, the equation of the line passing through them is:

$$\frac{y - (-2)}{-5 - (-2)} = \frac{x - 3}{4 - 3}$$

$$\frac{y + 2}{-3} = x - 3$$

$$y + 2 = -3(x - 3)$$

$$y + 2 = -3x + 9$$

$$y = -3x + 7$$

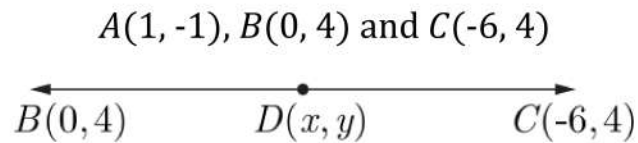
Since point $A(x, y)$, is also collinear with points B and C , it must satisfy this equation.

Thus, option (d) is the correct answer.

3.

Ans:

We have



(a) Coordinates of D

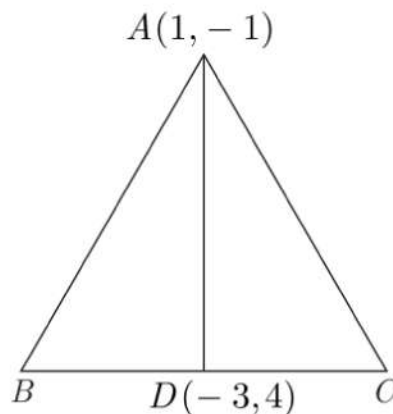
Since D is the mid-point of BC ,

$$D(x, y) = \left(\frac{0 + (-6)}{2}, \frac{4 + 4}{2} \right)$$

$$D = \left(\frac{-6}{2}, \frac{8}{2} \right)$$

$$D = (-3, 4)$$

Hence, the coordinates of D are $(-3, 4)$.



(b) Equation of the median AD

The median AD passes through points

$$A(1, -1) \text{ and } D(-3, 4)$$

Using the formula for the equation of a line through two points,

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$y - (-1) = \frac{4 - (-1)}{-3 - 1}(x - 1)$$

$$y + 1 = \frac{5}{-4}(x - 1)$$

$$-4(y + 1) = 5(x - 1)$$

$$-4y - 4 = 5x - 5$$

$$5x + 4y - 1 = 0$$

Final Answers:

- Coordinates of D : $(-3, 4)$
- Equation of the median AD : $5x + 4y - 1 = 0$

4.

Ans:

(a) Coordinates of A and B

From the graph,

$$A = (4, 0) \text{ and } B = (0, 4)$$

(b) Coordinates of point P

Given,

$$AP : PB = 3 : 1$$

Using the section formula,

$$P(x, y) = \left(\frac{3x_2 + 1x_1}{3 + 1}, \frac{3y_2 + 1y_1}{3 + 1} \right)$$

$$P = \left(\frac{3(0) + 1(4)}{4}, \frac{3(4) + 1(0)}{4} \right)$$

$$P = \left(\frac{4}{4}, \frac{12}{4} \right) = (1, 3)$$

Hence, the coordinates of P are $(1, 3)$.

(c) Equation of the line through P and perpendicular to AB

Slope of line AB ,

$$m_1 = \frac{4 - 0}{0 - 4} = -1$$

Slope of a line perpendicular to AB ,

$$m_2 = \frac{1}{m_1} = -1$$

Equation of the line passing through $P(1, 3)$,

$$y - 3 = -1(x - 1)$$

$$y - 3 = 1 - x$$

$$x + y - 4 = 0$$

Final Answers:

(a) $A(4, 0)$, $B(0, 4)$

(b) $P(1, 3)$

(c) Equation of the required line: $x + y - 4 = 0$

5.

Ans: (c) -2

Explanation:

Given equation of the line:

$$5x - 4y - 8 = 0$$

Rewriting it in the form $y = mx + c$:

$$-4y = -5x + 8$$

$$y = \frac{5}{4}x - 2$$

Comparing with the standard form $y = mx + c$, we get:

$$c = -2$$

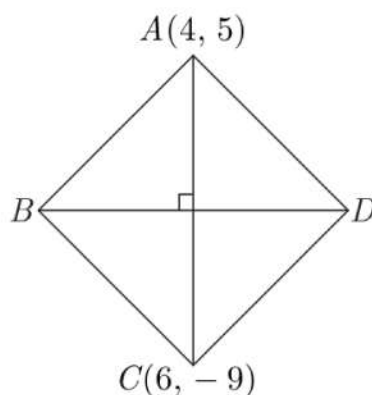
Thus, the y-intercept of the line is -2 .

Hence, option (c) is the correct answer.

6.

Ans:

As per given question we have drawn figure below.



Slope of AC $m_1 = \frac{5+9}{4-6} = \frac{14}{-2} = -7$

Since diagonals of a rhombus bisect each other at right angles, BD is perpendicular to AC .

Slope of BD $m_2 = -\frac{1}{m_1}$
 $= -\frac{1}{-7} = \frac{1}{7}$

Midpoint of AC $M = \left(\frac{4+6}{2}, \frac{5-9}{2} \right)$
 $= (5, -2)$

Mid point M is also mid point of diagonal of BD .

Thus, equation of line having slope $\frac{1}{7}$ and passing through $(5, -2)$,

$$y - y_1 = m(x - x_1)$$

$$y + 2 = \frac{1}{7}(x - 5)$$

$$7y + 14 = x - 5$$

$$7y = x - 19$$

$$x - 7y - 19 = 0$$